CHM2242 Organic Chemistry – Typed Lecture Notes

Oct 31, 2026 · @Abdullahi Idris

1. Alcohols

Structure, properties and reactions of alcohol (notes dated 28/09/26). R = aryl, alkyl or cyclic.

Course outline (as listed in the notes)

1. Alcohols, ethers, epoxides
2. Aldehydes and ketones
3. Carboxylic acids and their derivatives
4. Amines
5. Introduction to α,β-unsaturated carbonyl compounds

General formula and classification

Alcohols have the general formula **R–OH** (R = alkyl group).

Class	Structure	Note
Methyl alcohol	CH_3–OH	
1° alcohol	R–CH_2–OH	
2° alcohol	R_2CH–OH (R and R′ on the carbinol carbon)	
3° alcohol	R_3C–OH	

Classification examples drawn in the notes

- 1-Methylcyclopentanol (CH_3 and OH on the same ring carbon): 3°
- Cyclohexanol (ring carbon bearing H and OH): 2°
- Ring–CH_2OH (benzyl-type alcohol, drawn as a ring with CH_2OH): 1°
- Cyclopentanol: 2°

- CH_2=CH–OH: written as 2° in the notes (the carbon bearing OH is bonded to only one other carbon, so check this one against your lecturer's answer)
- C_3H…–$C(CH_3)(OH)$–CH_3: 3° (the alkyl chain is partly cut off in the notes)

Isomerism

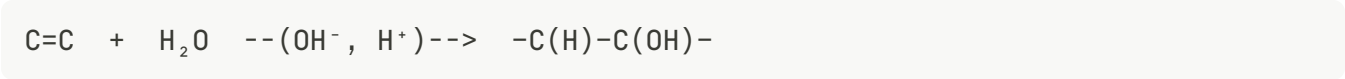
- C_2H_6O: CH_3CH_2OH (ethanol) and CH_3–O–CH_3 (dimethyl ether)
- C_3H_6O: CH_2=CH–CH_2–OH, CH_2=C(OH)–CH_3, and CH(OH)=CH–CH_3
- Also noted at the top of the next page (C_3H_6O isomers): cyclopropanol (a three-membered ring bearing OH), CH_3CH_2–CHO and CH_3–CO–CH_3

Physical properties

- Water and alcohols have similar physical properties.
- Alcohols are soluble in water.
- The presence of the –OH group makes them polar.
- The next point begins "Lo…" and is cut off at the bottom of the page (probably "Low…"; it is not finished in the notes).

Preparation of alcohols

1) Hydration of alkenes



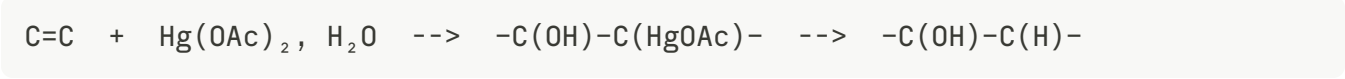
a. Acid-catalysed hydration of an alkene (involves protonation)



Mechanism:

1. Protonation of the double bond to form a carbocation: C=C + H–O⁺(H)–H → H–C–C⁺ + H_2O
2. Nucleophilic attack by water on the carbocation gives a protonated alcohol: –C⁺ + $H_2\ddot{O}$: → –C–O⁺(H)–H
3. Deprotonation gives the alcohol: –C(H)–C–$\ddot{O}^+H_2$ + H_2O ⇌ –C(H)–C(OH)– + H_3O^+

b. Hydration by oxymercuration–demercuration



Mercuric acetate: $Hg(OAc)_2$ = $Hg(O_2CCH_3)_2$. (The second-step reagent is not written on the page; it is the reduction that replaces HgOAc with H, $NaBH_4$ as in the ether notes.)

c. Hydroboration

```
CH3–C(CH3)=CH–CH3  --(B2H6)-->  CH3–CH(CH3)–CH(BH2)–CH3  (an alkylborane)  --(H2O2, OH⁻)-->  CH3–CH(CH3)–CH(OH)–CH3
```

Product: 3-methylbutan-2-ol.

2) Formation of alcohols from aldehydes and ketones using Grignard reagents (GR)

Grignard reagents: R–Mg–X, for example CH_3–Mg–Cl and C_2H_5–MgBr.

Carbonyl compound	Reaction	Product
Formaldehyde, H–CHO	H–C(=O)–H + R–MgX → …	R–CH_2–OH (1°)
Aldehyde (other than formaldehyde)	R–C(=O)–H + R′MgX → …	R–CH(R′)–OH (2°)
Ketone	R–C(=O)–R + R′MgX → …	R–C(R′)(R)–OH (3°)

Example: C_6H_5–MgBr + $(CH_3)_2C{=}O$, in ether → C_6H_5–$C(CH_3)_2$–OH

Exercise given in the notes: show how the following can be prepared

1. Cyclohexyl–CH_2OH (hexagon ring bearing CH_2OH)
2. Cyclopentyl–CH_2OH (pentagon ring bearing CH_2OH)
3. A branched primary alcohol drawn as a skeleton: a methyl-branched chain ending in –OH
4. Cyclopentyl–$C(CH_3)(OH)$–CH_2OH (ring carbon attached to a carbon bearing CH_3, OH and CH_2OH)
5. 3-Phenylhexan-3-ol

3) Reduction of carbonyl compounds with hydride reagents ($NaBH_4$, $LiAlH_4$)

```
C=O  --(NaBH4 or LiAlH4)-->  –CH–OH
```

4) Alcohols from hydrolysis of an ester

```
R–C(=O)–OR  +  H2O  -->  R–OH  +  R–COO⁻  (carboxylate)
```

Example (as written): $CH_3CH_2–C(=O)–CH_2CH_3 + Na^+OH^- \rightarrow CH_3CH_2–OH + CH_3CH_2–C(=O)–O^-Na^+$. The ester oxygen is missing from the starting material in the notes; it should read $CH_3CH_2–C(=O)–O–CH_2CH_3$ (ethyl propanoate). A margin note on the page says "ether" next to the ester.

2. Reactions of alcohols (R–OH)

Generally, R–OH undergo substitution (SN1, SN2), elimination (E1, E2), esterification, reduction and oxidation reactions, as well as formation of ethers.

Substitution reactions of R–OH

1. Conversion of R–OH to alkyl halides

The most commonly used reagents for this conversion are:

- Hydrogen halides (HCl, HBr, HI)
- Phosphorus tribromide (PBr_3)
- Thionyl chloride ($SOCl_2$)

a) Reaction of R–OH with HX

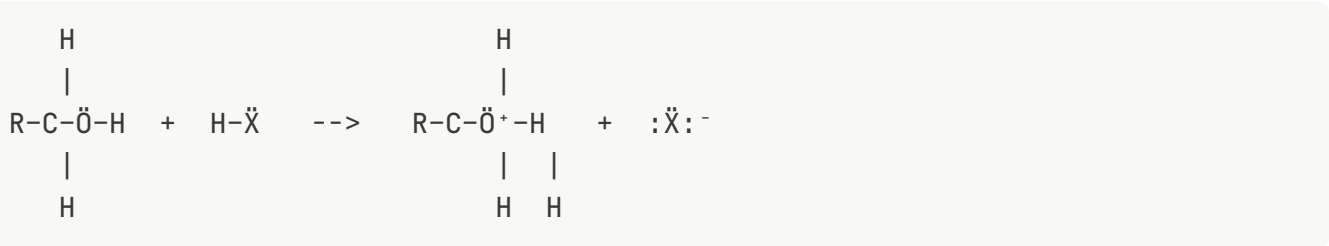
```
R–OH  +  HX  -->  R–X  +  H2O
```

Note: 1° alcohols and MeOH react to form alkyl halides under acidic conditions by the SN2 mechanism.

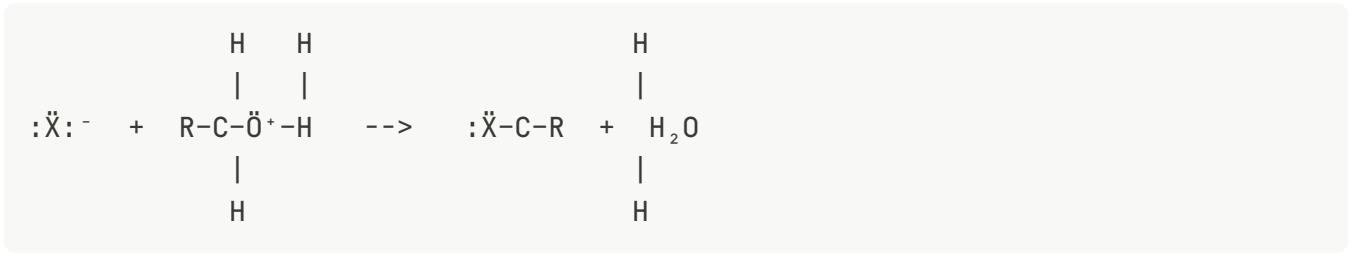
Example from the earlier notes (5/10/2025): $CH_3–CH(OH)–CH_2–CH_3 + HI \rightarrow CH_3–CH(I)–CH_2–CH_3 + H_2O$.

Mechanism:

- **Step 1: protonation of the alcohol by an acid.**

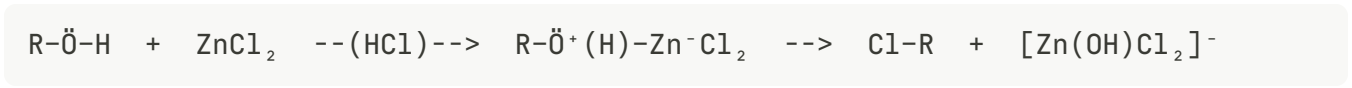


- **Step 2: displacement of the poor leaving group (H_2O) by the halide ion.**



where X = Br, I.

However, hydrogen chloride does not react with 1° or 2° alcohols unless $ZnCl_2$ or some other Lewis acid is added to the reaction mixture:



b) Reaction of R–OH with phosphorus tribromide (PBr_3)



Example: reaction of (R)-2-pentanol with PBr_3.

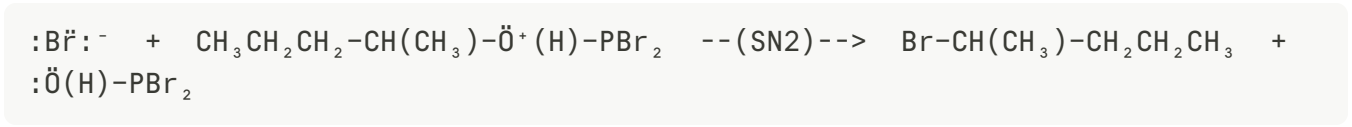


Mechanism:

- **Step 1: displacement of bromide and formation of the leaving group.** The oxygen lone pair attacks phosphorus and bromide leaves:



- **Step 2: bromide displaces the leaving group to give (S)-2-bromopentane** (SN2, back-side attack):

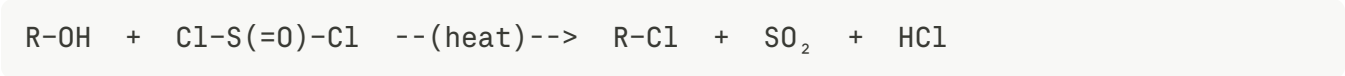


Notes:

1. Rearrangement is uncommon because no carbocation is formed.
2. This mechanism is SN2 displacement where Br^- attacks the back side of the alkyl group. The attack is hindered if the alkyl group is tertiary.

c) Reaction of R–OH with thionyl chloride ($SOCl_2$)

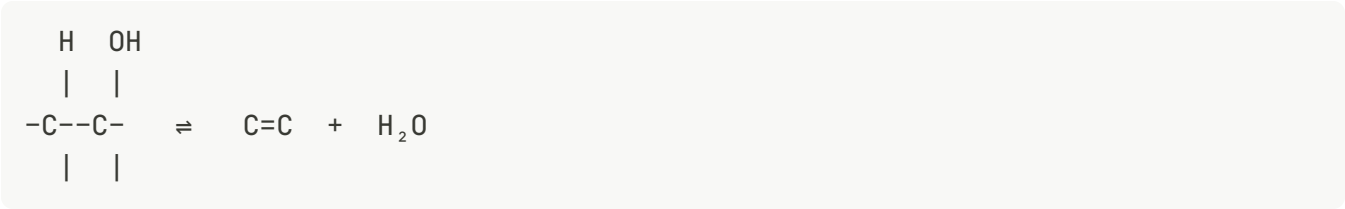
(The notes write the formula as $SOCl_3$ in this heading; it is $SOCl_2$, as written in the reagent list.)



Mechanism:

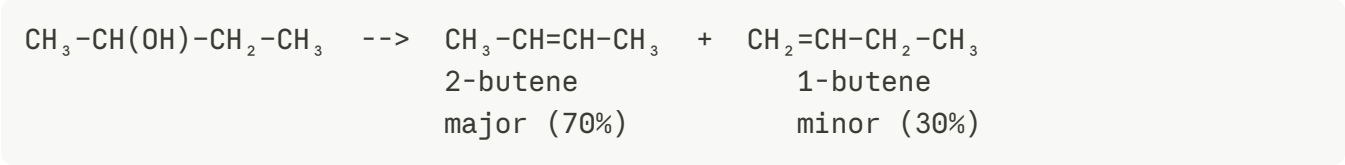
1. The alcohol oxygen attacks sulfur: R–Ö–H + $S(=O)Cl_2$ → R–Ö⁺(H)–S(=O)Cl + Cl^-
2. Loss of H^+ (as HCl) gives the alkyl chlorosulfite: R–O–S(=O)–Cl + HCl
3. Chloride attacks carbon in the fast step, releasing SO_2: R–O–S(=O)Cl → R–Cl + SO_2 (the notes draw it as R^+ and $^-$O–S(=O)Cl collapsing to R–Cl + S=O)

Dehydration of R–OH



Dehydration follows **Zaitsev's rule**: if two or more alkenes might be formed by deprotonation of the carbocation, the most highly substituted alkene predominates.

Example (2-butanol, dehydrated; the minor product is cut off at the page edge):



3. Ethers

Ethers are organic compounds of the general formula **R–O–R**, **Ar–O–R** or **Ar–O–Ar**, where Ar is phenyl or some other aromatic group.

Naming of ethers

1) Name the two groups attached to oxygen, and follow these names with the word "ether". Use alphabetical order if the two alkyl groups are not identical.

Structure (as drawn)	Name
$CH_3CH_2–O–CH_2CH_3$	Diethyl ether
$(CH_3)_2CH–O–CH(CH_3)_2$	Diisopropyl ether
$(CH_3)_3C–O–CH_3$	tert-Butyl methyl ether
$CH_3CH_2CH(CH_3)–O–CH_3$	2-Butyl methyl ether (sec-butyl methyl ether)

2) Name as alkoxy followed by the alkane name of the longest alkyl group.

Structure (as drawn)	Name
$CH_3CH_2–O–CH_2CH_3$	Ethoxyethane
$CH_3CH_2–O–CH_2CH_2CH_2CH_3$	Ethoxybutane
$CH_3CH_2CH(OCH_3)CH_2CH_2CH_3$	3-Methoxyhexane

Symmetrical ethers contain two identical alkyl groups, while unsymmetrical ethers have two different alkyl groups.

Preparation of ethers

1) Williamson synthesis: treatment of an alkyl halide with a sodium alkoxide.

```
R–X  +  R'O⁻Na⁺   -->  R–O–R'
R–X  +  ArO⁻Na⁺   -->  R–O–Ar
```

Yield from R–X: CH_3 > 1° > 2° > 3°.

Example: phenol + CH_3CH_2Br, aq. NaOH → ethoxybenzene ($C_6H_5–O–CH_2CH_3$)

2) Alkoxymercuration–demercuration

```
C=C  +  ROH  +  Hg(OOCCF₃)₂  -->  –C(OR)–C(HgOOCCF₃)–  --(NaBH₄)-->  –C(OR)–C(H)–
```

The reaction follows Markovnikov's rule.

Reactions of ethers

Ethers are very unreactive, hence they are used as solvents. However, they are cleaved by acids to form R–X.

Acid-catalysed cleavage of ethers

```
R–OH  +  HX  -->  R–X  +  H2O          (alcohol + hydrogen halide → alkyl
halide + water)
R–O–R'  +  HX  -->  R–X  +  R'–OH
```

The cleavage is carried out with excess hydrogen halide and heat:

```
R–O–R'  +  2 HX  --(heat)-->  R–X  +  R'–X  +  H2O

CH3CH(OCH3)CH2CH3  --(HBr, heat)-->  CH3CH(Br)CH2CH3  +  CH3Br
```

Reactivity of HX: **HI > HBr >> HCl**

Mechanism of the reaction

```
CH3CH2–O–CH2CH3  +  HBr  --(heat)-->  2 CH3CH2Br  +  H2O
```

1. **Protonation of oxygen (fast, reversible):**

```
CH3CH2–Ö–CH2CH3  +  H–Br   ⇌   CH3CH2–O+(H)–CH2CH3  +  Br-
                                  diethyloxonium ion
```

2. **Nucleophilic attack of the halide anion on a carbon of the dialkyloxonium ion (slow):**

```
Br-  +  CH3CH2–O+(H)–CH2CH3   --(slow)-->  CH3CH2Br  +  CH3CH2OH
```

3. **Third and fourth steps** (the alcohol formed is protonated, then attacked by bromide):

```
CH3CH2OH  +  HBr   ⇌ (fast)   CH3CH2–O+H2  +  Br-
Br-  +  CH3CH2–O+H2   --(slow)-->  CH3CH2Br  +  H2O
```

Crown ethers

Crown ethers are cyclic polyethers containing repeating –OCH_2CH_2– units. They look cumbersome but are actually pretty simple to name. They are named **x-crown-y**, where x is the total number of carbon and oxygen atoms in the cyclic backbone and y is the number of oxygen atoms.

Name	Ring atoms (x)	Oxygen atoms (y)
12-crown-4	12	4
14-crown-4	14	4
15-crown-5	15	5

Crown ethers have valuable synthetic advantages. They form stable metal complexes by binding to small metal cations such as K^+, Na^+, etc., using the oxygen atoms as donors. In other words, the metal cations are held in the cavity of the crown ether.

4. Epoxides

Epoxides are cyclic ethers with a three-membered ring. The structure of an epoxide has an oxygen atom attached to two adjacent carbon atoms.

```
      O
     / \          (drawn in the notes: the basic epoxide ring, a methyl-
substituted epoxide,
    C---C          and epoxides fused to cyclobutane and cyclohexane rings)
```

Naming of epoxides

- Consider the longest carbon chain as the parent name.
- Count from the side nearest to oxygen.
- Specify the positions of the carbon atoms bearing the oxygen using the numbers 1, 2, …
- Add the word "epoxy" after the numbers.

Structure	Name
Three-membered ring, O on C1 and C2 of a two-carbon chain	1,2-Epoxyethane
Epoxide on C2 and C3 of a four-carbon chain	2,3-Epoxybutane
Epoxide fused to a cyclobutane ring	1,2-Epoxycyclobutane

Preparation of epoxides

1) Ethylene oxide (1,2-epoxyethane) is mainly prepared by exposing an alkene (ethene) to oxygen in the presence of silver oxide as a catalyst at high temperature.

```
CH2=CH2  --(O2, Ag2O, 300 °C)-->  ethylene oxide (three-membered ring C–C–O)
```

2) Epoxides are prepared by treatment of an alkene with a peroxyacid (RCO_3H), typically m-chloroperoxybenzoic acid.

```
cyclohexene  +  m-chloroperoxybenzoic acid   --(CH2Cl2 solvent)-->  1,2-
epoxycyclohexane  +  m-chlorobenzoic acid
```

(m-Chloroperoxybenzoic acid: a benzene ring with Cl at the meta position and a –C(=O)–O–O–H group. m-Chlorobenzoic acid: the same ring with –C(=O)–OH.)

3) Treatment of a halohydrin with a base.

```
C=C  +  X2  --(H2O)-->  –C(OH)–C(X)–  +  HX        (X2 = halogen; product =
halohydrin)

–C(OH)–C(X)–  --(base)-->  epoxide  +  HX

CH3–CH(OH)–CH2Br  --(OH⁻)-->  methyl-epoxide (propylene oxide)  +  HBr
```

Mechanism: hydroxide removes the proton of the OH group; the resulting alkoxide oxygen attacks the neighbouring carbon from the back side and bromide leaves (an intramolecular SN2), closing the three-membered ring.

Reactions of epoxides: ring opening

Acid-catalysed epoxide opening

Epoxides are cleaved by treatment with acid just as other ethers are, but under milder conditions because of the strong ring strain.

```
epoxide  --(H⁺, Nu–H)-->  H–C(H)(Nu⁺H)–C(H)(H)–OH  --(–H⁺)-->  Nu–CH2–CH2–OH
```

Mechanism:

1. Protonation of the epoxide oxygen by H^+.
2. Nu–H attacks a ring carbon from the back side, opening the ring and giving $Nu^+(H)$–C–C–OH.

3. Loss of H^+ (taken by another Nu–H, which becomes Nu^+H_2) gives Nu–C–C–OH.

Examples:

```
1,2-epoxycyclohexane  --(H3O+)-->  protonated epoxide  --(H2O attacks from
the back)-->  cyclohexane-1,2-diol (trans)  +  H3O+

1,2-epoxycyclohexane  --(HX, anhydrous, ether)-->  2-halocyclohexanol   (OH
on one carbon, X on the adjacent carbon)
```

Regiochemistry

- When both epoxide carbon atoms are either 1° or 2°, the nucleophile attacks mainly the less highly substituted site, hence an **SN2-like** result.
- When one of the epoxide carbon atoms is tertiary, nucleophilic attack occurs primarily at the more highly substituted site, hence an **SN1-like** result.

Examples drawn in the notes (H^+, then CH_3OH, then $-H^+$). The products below follow the rule stated in the notes, because the skeleton drawings are small:

```
1°/2° epoxide (propylene oxide):   CH3–CH(–O–)CH2  --(H+, CH3OH)-->  CH3–
CH(OH)–CH2–OCH3
                                    (attack at the less substituted CH2)

3° epoxide (2,2-dimethyloxirane):  (CH3)2C(–O–)CH2  --(H+, CH3OH)-->  HO–CH2–
C(CH3)2–OCH3
                                    (attack at the tertiary carbon)
```

Base-catalysed epoxide opening

The strain of the epoxide ring causes it to react with hydroxide ion.

```
cyclohexane ring carbon bearing the epoxide (C–O–CH2)  --(OH-)-->  ring
C(O-)–CH2–OH  --(H2O)-->  ring C(OH)–CH2–OH
```

They can react with Grignard reagents as well:

```
R–MgBr  +  ethylene oxide  -->  R–CH2CH2–O- +MgBr  --(H3O+)-->  R–CH2CH2–OH
```

Base-catalysed epoxide opening is typical of an **SN2 reaction**, in which attack of the nucleophile takes place at the less hindered epoxide carbon atom.

```
epoxide on a cyclohexane ring carbon  --(CH3CH2OH / -OCH2CH3)-->  ring C(OH)-CH2-OCH2CH3
```

The ethoxide attacks the CH_2 carbon (the less hindered one), so the OH stays on the ring carbon.

5. Aldehydes and ketones

Preparation, properties and reaction mechanisms of: (1) aldehydes, (2) ketones, (3) α,β-unsaturated carbonyl compounds.

```
   O                  O
   ||                 ||
R-C-H              R-C-R'
aldehyde            ketone
```

Nomenclature of aldehydes

The IUPAC name of an aldehyde is obtained by replacing the final "e" in the name of the corresponding alkane with **"al"**.

Structure	Name
H–CHO	Methanal
CH_3–CHO	Ethanal (acetaldehyde)
$CH_3CH_2CH_2$–CHO	Butanal (butyraldehyde)
CH_3CH_2–CH(Cl)–CHO	2-Chlorobutanal
$(CH_3)_2$CH–CHO	2-Methylpropanal
OHC–$CH_2CH_2CH_2$–CHO	Pentanedial

If the aldehyde is attached to a ring, it is named by adding **carbaldehyde** to the name of the ring:

- Cyclohexane ring bearing –CHO: cyclohexanecarbaldehyde
- Benzene ring bearing –CHO: benzenecarbaldehyde (benzaldehyde)

Nomenclature of ketones

The systematic name of a ketone is obtained by replacing the "e" at the end of the name of the corresponding alkane with **"one"**. The carbon atoms are numbered from the end that gives the carbonyl carbon the smaller number.

Structure	Name
$CH_3–CO–CH_3$	Propanone
$CH_3CH_2–CO–CH_3$	Butanone (2-butanone)
Skeleton drawing, 6-carbon chain with a methyl branch	5-Methyl-2-hexanone
Six-membered ring with C=O	Cyclohexanone
$CH_3–CO–CH_2–CO–CH_3$	2,4-Pentanedione
$C_6H_5–CO–CH_3$	Methyl phenyl ketone / phenylethanone (acetophenone)

Relative reactivities of carbonyl compounds

The carbonyl group is polar because oxygen is more electronegative than carbon. Thus the carbonyl carbon is electron deficient and carries a partial positive charge (δ+). It is therefore an **electrophile** and reacts with nucleophiles.

```
   Oδ⁻
   ‖
R–Cδ⁺–R(H)    ←  Nu:⁻ attacks the carbonyl carbon
```

An aldehyde has a greater δ+ on its carbonyl carbon than a ketone, because H is more electron-withdrawing than alkyl.

```
Most reactive:  H–C(=O)–H   >   R–C(=O)–H   >   R–C(=O)–R'   :least reactive
```

Reactions of aldehydes and ketones

Aldehydes and ketones react with nucleophiles to form addition products.

```
R–C(=O)–R'  +  Z:⁻   -->   R–C(O⁻)(Z)–R'   --(H₃O⁺)-->   R–C(OH)(Z)–R'
               (strong Nu)
```

- The reaction above is an example of an **irreversible nucleophilic addition**, because the nucleophile is a strong base and the tetrahedral compound does not have a group that can be eliminated.
- On the other hand, if the nucleophile is a weak base, the reaction will be a **reversible nucleophilic addition**, because the tetrahedral compound can eliminate the weak base and revert to the starting material (Z can be eliminated).

```
R–C(=O)–R'  +  Z:⁻   ⇌   R–C(O⁻)(Z)–R'   ⇌ (H₃O⁺)   R–C(OH)(Z)–R'       (weak Nu, reversible)
```

- If the attacking atom of the nucleophile has a lone pair and there is enough acid in the solution to protonate the –OH group, water can be eliminated from the addition product. Thus the fate of the tetrahedral product depends on the identity of Z. This is called a **nucleophilic addition–elimination reaction**.

```
R–C(OH)(:Z)–R'   --(H₃O⁺)-->   R–C(O⁺H₂)(:Z)–R'   ⇌   R–C(=Z⁺)–R'  +  H₂O
```

1. Reaction with Grignard reagents

```
R–C(=O)–R'  +  R''–MgBr   -->   R–C(O⁻ ⁺MgBr)(R'')–R'   --(H₃O⁺)-->   R–C(OH)(R'')–R'
```

- **Formaldehyde gives a primary alcohol:**

```
H–C(=O)–H  +  CH₃CH₂–MgBr   -->   H–C(H)(O⁻ ⁺MgBr)–CH₂CH₃   --(H₃O⁺)-->   CH₃CH₂CH₂OH
```

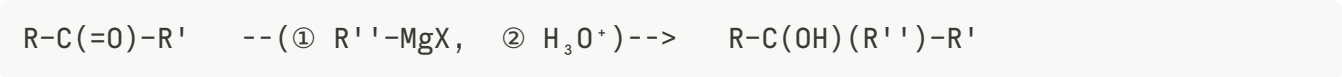
- **An aldehyde other than formaldehyde gives a secondary (2°) alcohol:**

```
CH₃CH₂–C(=O)–H  +  CH₃CH₂–MgBr   -->   CH₃CH₂–CH(O⁻ ⁺MgBr)–CH₂CH₃   --(H₃O⁺)-->   CH₃CH₂–CH(OH)–CH₂CH₃
(propanal)
```

- **A ketone gives a tertiary alcohol:**

```
CH₃–C(=O)–CH₂CH₃  +  CH₃CH₂–MgBr   -->   CH₃–C(O⁻ ⁺MgBr)(CH₂CH₃)–CH₂CH₃   --(H₃O⁺)-->   CH₃–C(OH)(CH₂CH₃)₂
```

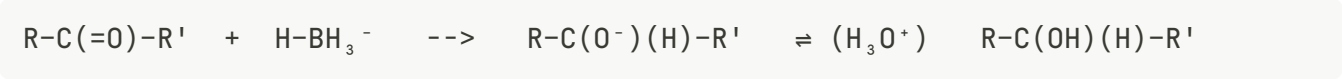
The reaction can generally be written as:



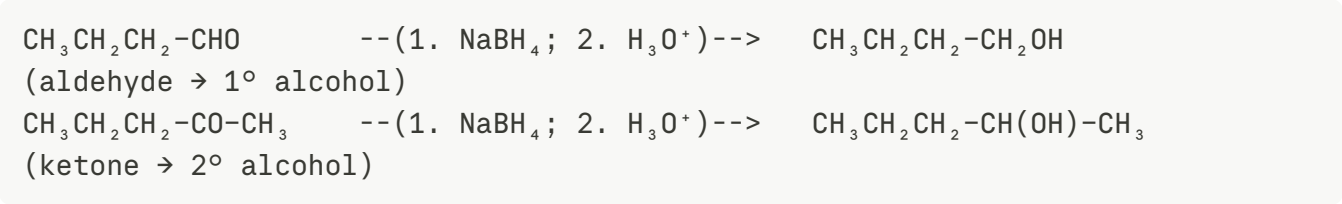
The order of numbering means that the acid is added after the Grignard reagent has reacted.

2. Reaction with hydride ion

Aldehydes and ketones react with a hydride ion to form a nucleophilic addition product.

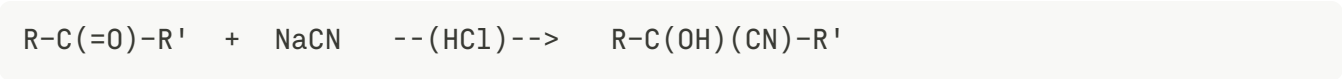


In the presence of a hydride source (such as $NaBH_4$), aldehydes and ketones are generally reduced to primary and secondary alcohols respectively:

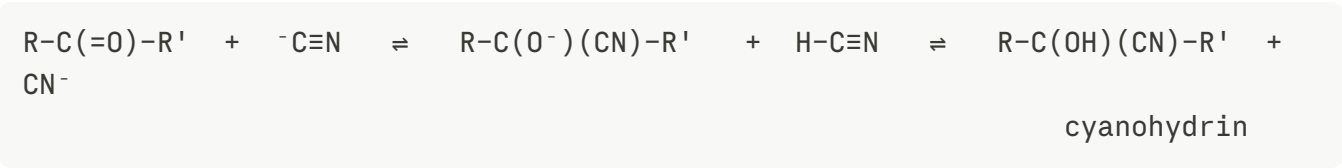


3. Reaction with hydrogen cyanide

An aldehyde or ketone reacts with HCN to form a **cyanohydrin**.



Mechanism:



In basic solution, a cyanohydrin is converted back to the carbonyl compound, for example cyclohexanone cyanohydrin ⇌ (HO^-/H_2O) ⇌ alkoxide ⇌ cyclohexanone + CN^-.

4. Reaction with amines and amine derivatives

An aldehyde or a ketone reacts with a primary amine to form an **imine (Schiff base)**.

```
C=O  +  R–NH2   ⇌ (H+)   C=N–R  +  H2O
                          imine (Schiff base)

R–C(=O)–H  +  R–NH2   ⇌ (H+)   R–CH=N–R      aldehyde + 1° amine → imine
R2C=O      +  R–NH2   ⇌ (trace H+)   R2C=N–R   ketone + 1° amine → imine

e.g.  R2C=O  +  C6H5–CH2–NH2   ⇌ (trace H+)   R2C=N–CH2–C6H5  +  H2O
```

Mechanism (B: = any species that can remove a proton, H–B⁺ = any species that can donate a proton):

1. The amine nitrogen attacks the carbonyl carbon: R(R′)C(O⁻)–N⁺H₂R″.
2. Proton transfers give the carbinolamine R(R′)C(OH)–NHR″.
3. The –OH is protonated (H–B⁺), giving R(R′)C(O⁺H₂)–NHR″.
4. Loss of water gives the iminium ion R(R′)C=N⁺HR″.
5. Deprotonation by B: gives the imine R(R′)C=N–R″ (and H–B⁺).

In acidic medium, the imine is hydrolysed back to the carbonyl compound:

```
R2C=N–R''  +  H2O   --(HCl)-->   R2C=O  +  R''–NH3+
```

Formation of imine derivatives

- An aldehyde or a ketone reacts with hydroxylamine to give an **oxime**:

```
R2C=O  +  H2N–OH   ⇌ (trace H+)   R2C=N–OH
                                    oxime
```

- They also react with hydrazine to give a **hydrazone**:

```
R2C=O  +  H2N–NH2   ⇌ (H+)   R2C=N–NH2
                               hydrazone
```

- With 2,4-dinitrophenylhydrazine:

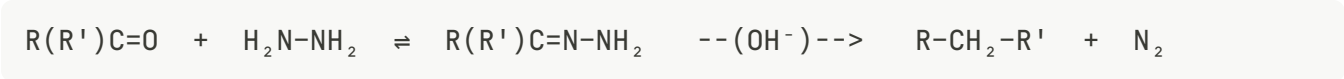
```
R2C=O  +  H2N–NH–C6H3(NO2)2   ⇌ (trace H+)   R2C=N–NH–C6H3(NO2)2
          2,4-dinitrophenylhydrazine              2,4-dinitrophenylhydrazone
```

- *(Boxed and marked "Ignore" in your notes.)* Aldehydes and ketones react with semicarbazide to give a semicarbazone:

```
R2C=O  +  H2N–NH–C(=O)–NH2   ⇌ (H+)   R2C=N–NH–C(=O)–NH2  +  H2O
```

Wolff–Kishner reduction

An aldehyde or ketone reacts with hydrazine to give a hydrazone. In the presence of a base, the reaction continues to form an alkane.

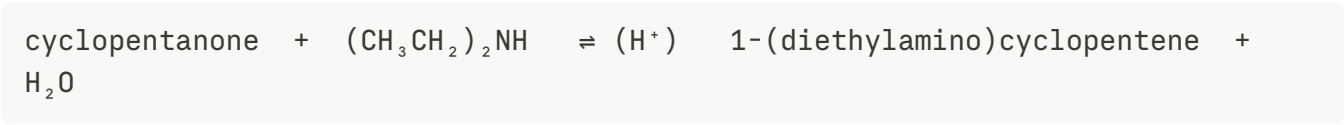


Mechanism:

1. Hydroxide removes an N–H proton of the hydrazone, giving R(R′)C=N–N⁻–H ↔ R(R′)C⁻–N=N–H.
2. The carbanion is protonated by water: R(R′)CH–N=N–H (+ OH^-).
3. Hydroxide removes the remaining N–H proton: R(R′)CH–N=N^- (+ H_2O).
4. Loss of N_2 gives the carbanion R(R′)CH^-, which is protonated by water: R–CH_2–R′ + OH^- (+ N_2).

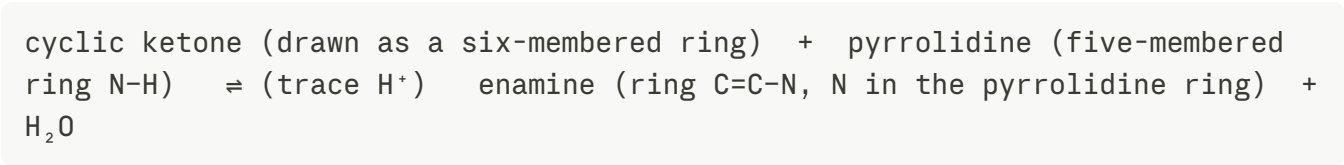
Enamines

Aldehydes and ketones react with secondary (2°) amines to form enamines.



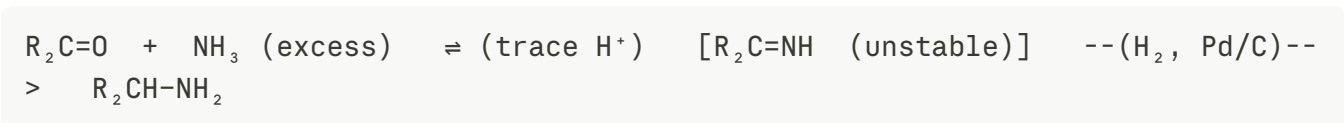
The mechanism is similar to that of imine formation: nucleophilic attack of the amine, proton transfers to the carbinolamine, protonation of –OH and loss of water give the iminium ion R_2N^+=C. **The intermediate cannot lose a proton from N, so it loses an H^+ from an α-carbon** (to B:), which gives the C=C of the enamine.

Exercise in the notes: propose a mechanism for



Reductive amination

An imine formed from the reaction of an aldehyde or ketone with NH_3 is relatively unstable. It serves as a useful intermediate. If the reaction with ammonia is carried out in the presence of H_2 and an appropriate metal catalyst, hydrogen adds to the C=N bond as it is formed, giving a primary amine. This reaction is called **reductive amination**.



Secondary and tertiary amines can be prepared from imines and enamines using sodium cyanoborohydride ($NaBH_3CN$). This is called the **Borch reaction**.

```
cyclohexanone ring C=O  +  R–NH2   ⇌ (trace H⁺)   ring C=N–R (imine)   --
(NaBH3CN)-->   ring CH–NHR   (2° amine)

ring C=O  +  R2NH   ⇌ (trace H⁺)   ring C=C–NR2 (enamine, drawn as ring–NR2)
--(NaBH3CN)-->   ring CH–NR2   (3° amine)
```

5. Reaction with water

Aldehydes and ketones react with water to form **hydrates (gem-diols)**.

```
R–C(=O)–R(H)  +  H2O   ⇌ (HCl)   R–C(OH)2–R(H)
```

Mechanism (acid-catalysed):

1. $H_2O + H^+ \rightleftharpoons H_3O^+$, and the carbonyl oxygen is protonated: $R–C(=O^+H)–R'$.
2. Water attacks the carbonyl carbon: $R–C(OH)(O^+H_2)–R'$.
3. Deprotonation by water gives the hydrate $R–C(OH)_2–R' + H_3O^+$.

Reactions of aldehydes and ketones with alcohols

Addition of 1 equivalent of alcohol to an aldehyde yields a **hemiacetal**. Addition of a 2nd equivalent of alcohol yields an **acetal**. Like water, an alcohol is a poor nucleophile, so an acid catalyst is required for the reaction to take place at a reasonable rate.

```
R–C(=O)–H  +  CH3OH   ⇌ (HCl)   R–CH(OH)(OCH3)   ⇌ (CH3OH, HCl)   R–CH(OCH3)2
+  H2O
 aldehyde                          hemiacetal                          acetal
```

In the case of ketones, the addition products are called a **hemiketal** and a **ketal** respectively:

```
R–C(=O)–R  +  CH3OH   ⇌ (HCl)   R–C(OH)(OCH3)–R   ⇌ (CH3OH, HCl)   R–
C(OCH3)2–R
                                  hemiketal                           ketal
```

Mechanism:

1. The carbonyl oxygen is protonated ($H–B^+$).
2. CH_3OH attacks the carbonyl carbon.

3. Deprotonation gives R–C(OH)(OCH_3)–R′: a **hemiacetal** if R′ = H, a **hemiketal** if R′ = alkyl.
4. The –OH is protonated and water leaves, giving R–C(=O^+CH_3)–R′.
5. A second CH_3OH attacks, and deprotonation gives R–C(OCH_3)$_2$–R′: an **acetal** if R′ = H, a **ketal** if R′ = alkyl.

The Wittig reaction

An aldehyde or a ketone reacts with a phosphonium ylide to form an alkene.

```
R(R')C=O  +  (C6H5)3P=CH-CH3   -->   R(R')C=CH-CH3  +  (C6H5)3P=O
             phosphonium ylide                          triphenylphosphine
oxide
             (R' = H, alkyl)
```

Mechanism: the ylide carbon attacks the carbonyl carbon (O^- and $P^+(C_6H_5)_3$ on adjacent atoms, a four-membered O–P ring intermediate), then the ring collapses to give the alkene and O=P(C_6H_5)$_3$.

Examples:

```
(CH3CH2)2C=O  +  (C6H5)3P=CH-CH2CH3   -->   (CH3CH2)2C=CH-CH2CH3  (3-
ethylhex-3-ene)  +  (C6H5)3P=O

cyclic ketone (ring as drawn)  +  (C6H5)3P=CH2   -->   ring C=CH2  +
(C6H5)3P=O
```

6. Reactions at the α-carbon

Aldehydes and ketones have a second site for reactivity, i.e. the carbon bonded directly to the carbonyl carbon (the **α-carbon**). The hydrogen attached to the α-carbon (the **α-hydrogen**) is acidic enough to be removed by a strong base (the notes read "difficiently acidic", which I take to mean "sufficiently acidic").

```
      H   O                          O
      |   ||                         ||
R – C – C – R   +  :B   ⇌   R – C⁻H – C – R   +  H–B
```

Keto–enol interconversion: enolate formation

```
R–CH2–C(=O)–R   +  HO⁻   ⇌   R–CH⁻–C(=O)–R   ↔   R–CH=C(O⁻)–R   (enolate ion)
  keto tautomer

R–CH=C(O⁻)–R   +  H2O   ⇌   R–CH=C(OH)–R   +  OH⁻
                              enol tautomer
```

Halogenation of the α-carbon of aldehydes and ketones

Acid-catalysed halogenation (the notes say: no need for the mechanism)

```
cyclic ketone (ring C=O)  +  Cl2   --(H3O⁺)-->   α-chloro ketone (Cl on the
ring carbon next to C=O)  +  HCl

ketone  +  I2   --(H3O⁺)-->   α-iodo ketone  +  HI
```

Base-promoted halogenation (marked "Not needed" in the notes)

```
R–CH2–C(=O)–R  +  Br2 (excess)   --(HO⁻)-->   R–C(Br)2–C(=O)–R  +  2 Br⁻
```

Alkylation of the α-carbon of carbonyl compounds

```
cyclopentanone   --(LDA / THF)-->   enolate (O⁻ and ring C=C)   --(CH3CH2Br)-
->   2-ethylcyclopentanone  +  Br⁻
```

LDA = lithium diisopropylamide, $(CH_3)_2CH–N^-(Li^+)–CH(CH_3)_2$.

The aldol reaction

An **aldol addition** is a reaction in which one molecule of a carbonyl compound, after a proton is removed from an α-carbon, reacts as a nucleophile and adds to the electrophilic carbonyl carbon of a second carbonyl compound.

```
R–CH2–C(=O)–R      (electrophile)
R–C⁻H–C(=O)–R      (nucleophile)
```

Examples:

2 CH_3CH_2–CHO	⇌ (HO^-, H_2O)	CH_3CH_2–CH(OH)–$CH(CH_3)$–CHO	β-hydroxyaldehyde
2 CH_3–CO–CH_3	⇌ (HO^-, H_2O)	CH_3–$C(OH)(CH_3)$–CH_2–CO–CH_3	β-hydroxyketone

Mechanism (aldehyde):

1. HO^- removes an α-H: R–CH_2–CHO + HO^- ⇌ R–CH^-–CHO ↔ R–CH=CH–O^- + H_2O
2. The enolate attacks the carbonyl carbon of a second aldehyde molecule: R–CH_2–$CH(O^-)$–CHR–CHO
3. Protonation by water gives the β-hydroxyaldehyde R–CH_2–CH(OH)–CHR–CHO (and HO^- is regenerated).

Mechanism (acetone):

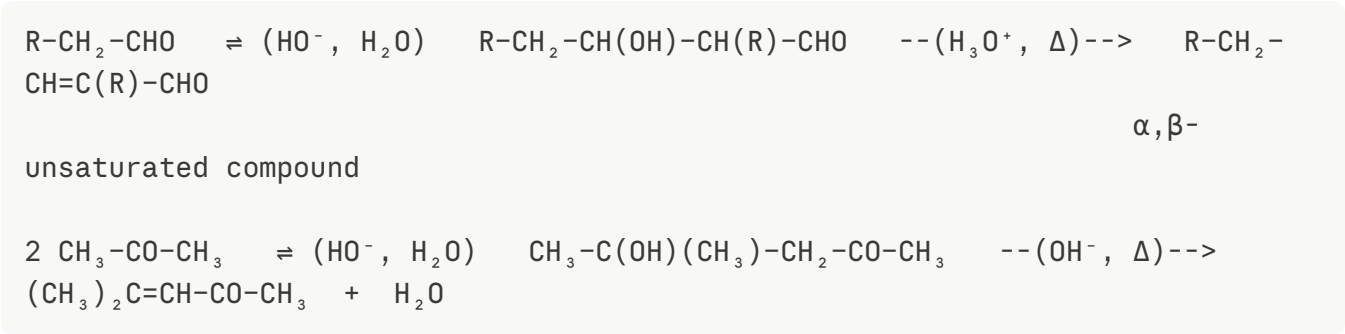
1. HO^- removes an α-H: CH_3–CO–CH_3 ⇌ CH_3–CO–CH_2^- ↔ CH_3–$C(O^-)$=CH_2
2. The enolate attacks a second acetone: CH_3–CO–CH_2–$C(CH_3)_2$–O^-
3. Protonation by water gives CH_3–CO–CH_2–$C(OH)(CH_3)_2$ (β-hydroxyketone) + HO^-.

Problems in the notes

1. Show the aldol addition product that would be formed from each of the following compounds:
   - (i) CH_3CH_2–CO–CH_2CH_3
   - (ii) cyclopentanone
   - (iii) $CH_3CH_2CH_2CH_2$–CHO
2. For each of the following compounds, indicate the aldehyde or ketone from which it would be formed by an aldol addition:
   - (i) 4-hydroxy-4-methyl-2-pentanone
   - (ii) 2,4-dicyclohexyl-3-hydroxybutanal

Aldol condensation

If the product of an aldol addition undergoes dehydration, the overall reaction is called an **aldol condensation**. An aldol condensation yields α,β-unsaturated carbonyl compounds.

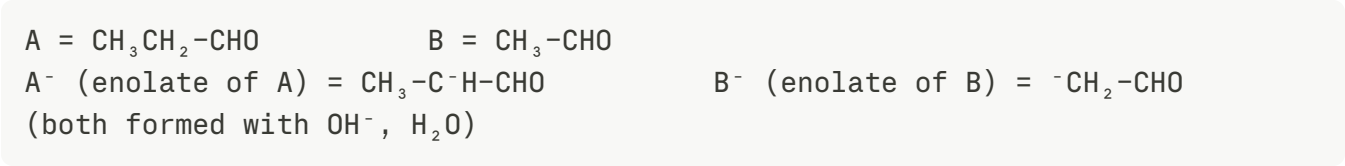


Mechanism of the dehydration:

1. A base (B:) removes the α-H that lies between the carbinol carbon and the C=O: R–CH(OH)–CH(H)–CHO.
2. This gives the enolate R–CH(OH)–CH=C(O^-)H.
3. The enolate expels OH^- from the β-carbon: R–CH=CH–CHO.

The crossed aldol addition

If two different carbonyl compounds are used in an aldol reaction, **four products** are obtained. Example: propanal (A) + ethanal (B).



Enolate	Attacks	Product
A^-	A	① CH_3CH_2–CH(OH)–CH(CH_3)–CHO
A^-	B	② CH_3–CH(OH)–CH(CH_3)–CHO
B^-	A	③ CH_3CH_2–CH(OH)–CH_2–CHO
B^-	B	④ CH_3–CH(OH)–CH_2–CHO

The four different products have similar physical properties, making them difficult to separate.

Directed aldol using LDA

On the other hand, LDA is used to remove the α-hydrogen that creates the enolate ion. Because LDA is a strong base, all the carbonyl compound will be converted to the enolate ion. An aldol addition will not be able to occur until the second carbonyl compound is

added. If the second carbonyl compound is added slowly, the chance that it will form an enolate ion and then react with another molecule of its parent compound will be minimized.

```
cyclopentanone  --(LDA)-->  enolate  --(① CH3CH2CH2–CHO added slowly;  ② H2O)-->  2-(1-hydroxybutyl)cyclopentanone

(ring C–CH(OH)–CH2CH2CH3)
```

7. α,β-Unsaturated aldehydes and ketones: nucleophilic addition

The resonance contributors of an α,β-unsaturated carbonyl compound show that the molecule has **two electrophilic sites: the carbonyl carbon and the β-carbon**.

```
  β   α   O                  O⁻                       O⁻
  R–CH=CH–C–R   ↔   R–CH=CH–C⁺–R   ↔   R–CH⁺–CH=C–R
```

Thus a nucleophile can add either to the carbonyl carbon or to the β-carbon.

Direct addition (1,2-addition)

Nucleophilic addition to the carbonyl carbon is called direct addition (1,2-addition).

```
Nu⁻  +  R–CH=CH–C(=O)–R   -->   R–CH=CH–C(O⁻)(Nu)–R   --(H3O⁺)-->   R–CH=CH–C(OH)(Nu)–R
```

Conjugate addition (1,4-addition)

Nucleophilic addition to the β-carbon is called conjugate addition or 1,4-addition, because it occurs at the 1 and 4 positions (numbering O = 1, C=O carbon = 2, α-carbon = 3, β-carbon = 4).

```
Nu⁻ + R–CH=CH–C(=O)–R   -->   [ R–CH(Nu)–CH⁻–C(=O)–R  ↔  R–CH(Nu)–CH=C(O⁻)–R ]   (resonance contributors)
        4    3   2 1

   -->   R–CH(Nu)–CH2–C(=O)–R     ⇌     R–CH(Nu)–CH=C(OH)–R
         keto tautomer                    enol tautomer
```

Which one forms?

- **If the nucleophile is a weak base, the conjugate addition product is always formed.** (Margin note in the notes: strong acids have weak conjugate bases.)

```
cyclohexenone  +  ⁻C≡N   --(HCl)-->   3-cyanocyclohexanone       (CN on the
β-carbon of the ring)

CH2=CH-CO-CH3  +  HBr   -->   Br-CH2-CH2-CO-CH3

CH3CH=CH-CO-CH3  +  (CH3)2NH   -->   CH3-CH(N(CH3)2)-CH2-CO-CH3
```

- **Strong bases as nucleophiles add directly to the carbonyl carbon.**

```
cyclohexenone  --(1. CH3MgBr;  2. H3O⁺)-->   1-methylcyclohex-2-en-1-ol
(OH and CH3 on the former carbonyl carbon; C=C kept)
```

A second example drawn beside this one is struck out in the notes.

8. Carboxylic acids and their derivatives: carboxylic acids

Carboxylic acids are organic compounds that contain a carbonyl group attached to a hydroxyl group. They are characterised by having one of the following structures:

```
  H-C(=O)-OH        R-C(=O)-OH        Ar-C(=O)-OH
```

Nomenclature

In systematic nomenclature, a carboxylic acid is named by replacing the terminal "e" of the alkane with **"oic acid"**.

Structure	Systematic name	Common name
H–COOH	Methanoic acid	Formic acid
CH_3–COOH	Ethanoic acid	Acetic acid
C_2H_5–COOH	Propanoic acid	Propionic acid
C_4H_9–COOH	Pentanoic acid	Valeric acid
C_5H_{11}–COOH	Hexanoic acid	Caproic acid

Structure	Systematic name	Common name
CH_2=CH–COOH	Propenoic acid	Acrylic acid
C_6H_5–COOH	Benzenecarboxylic acid	Benzoic acid

- In systematic nomenclature, the position of a substituent is designated by a number. The carbonyl carbon is always the C-1 carbon.
- In common nomenclature, the position of a substituent is designated by a lowercase Greek letter, and the carbonyl carbon is not given a designation.

```
CH3–CH2–CH2–CH2–CH2–C(=O)OH          CH3–CH2–CH2–CH2–CH2–C(=O)OH
    6    5    4    3    2    1              ε    δ    γ    β    α
(systematic nomenclature)                    (common nomenclature)
```

Examples:

```
CH3–CH2–CH(OCH3)–COOH      2-methoxybutanoic acid   (α-methoxybutyric acid)
CH3–CH2–CH(Br)–CH2–COOH    3-bromopentanoic acid    (β-bromovaleric acid)
```

Physical properties

- They are polar compounds and, like alcohols, can form hydrogen bonds with each other and with other kinds of molecules.
- The aliphatic acids show very much the same solubility behaviour as the alcohols: the first 4 members are miscible with water, the 5-carbon acid is partly soluble, and the higher acids are virtually insoluble.
- They have a higher boiling point than alcohols of comparable molecular weight, due to double hydrogen bonding.
- They are characterised by sharp, irritating and unpleasant odours.
- They react with alkali to form salts and water.

Occurrence and uses: acetic acid is in vinegar and other foods, and is also used in industry as a solvent. Fatty acids come from fats and oils. Benzoic acid is found in drugs and preservatives. Phthalic acid is used to make polyester.

Chemical properties (reactions)

i) Acidity / salt formation

```
R–COOH   ⇌   R–COO⁻  +  H⁺

2 CH₃COOH  +  Zn   -->   (CH₃COO)₂Zn  +  H₂↑            zinc acetate
CH₃CH₂–COOH  +  NaOH   -->   CH₃CH₂COONa  +  H₂O        sodium propanoate
C₆H₅–CO₂H  +  NaHCO₃   -->   C₆H₅–CO₂Na  +  CO₂↑  +  H₂O     sodium benzoate
```

ii) Conversion into functional derivatives

```
R–C(=O)–OH   -->   R–C(=O)–Z        (Z = Cl, OR, NH₂)
```

Examples (n = 1, 2 or 3, matching the stoichiometry):

```
n CH₃COOH  --(SOCl₂)-->  CH₃COCl  +  SO₂  +  HCl
n CH₃COOH  --(PCl₃)--->  CH₃COCl  +  H₃PO₃
n CH₃COOH  --(PCl₅)--->  CH₃COCl  +  POCl₃  +  HCl
                         (acid chloride)
```

(The PCl_3 line reads "CH_2COCl" in the notes; it should be CH_3COCl.)

Esterification and amide formation

```
CH₃CH₂–COOH  +  CH₃–OH   ⇌ (H⁺)   CH₃CH₂–C(=O)–OCH₃      (ester)

CH₃CH₂–COOH  --(SOCl₂)-->  CH₃CH₂–COCl  +  CH₃–OH  -->  CH₃CH₂–C(=O)–OCH₃
(ester)

CH₃CH₂–COOH  --(SOCl₂)-->  CH₃CH₂–COCl  --(NH₃)-->  CH₃CH₂–C(=O)–NH₂
(amide)
```

Order of reactivity of the alcohol: CH_3OH > 1° > 2° > 3°.

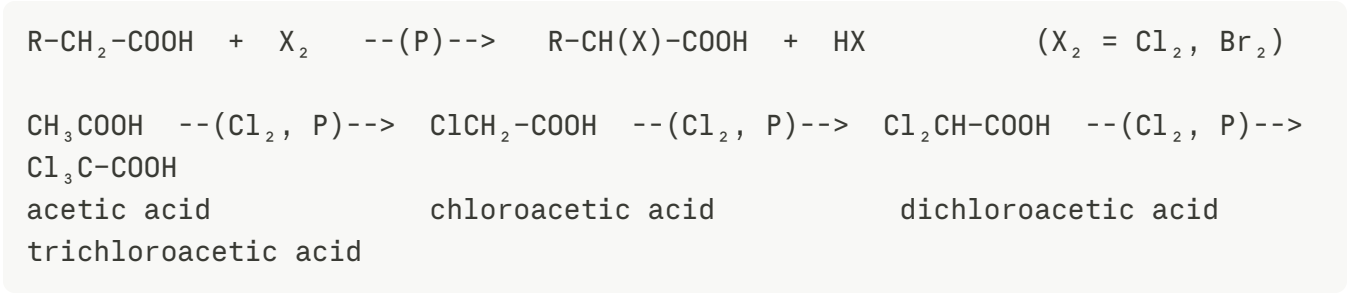
iii) Reduction

```
R–COOH  --(LiAlH₄)-->  R–CH₂–OH          (1° alcohol)

o-toluic acid (2-methylbenzoic acid)  --(LiAlH₄)-->  o-methylbenzyl alcohol
(2-methylphenyl–CH₂OH)
```

iv) Substitution in the alkyl or aryl group

(a) α-Halogenation of aliphatic acids:

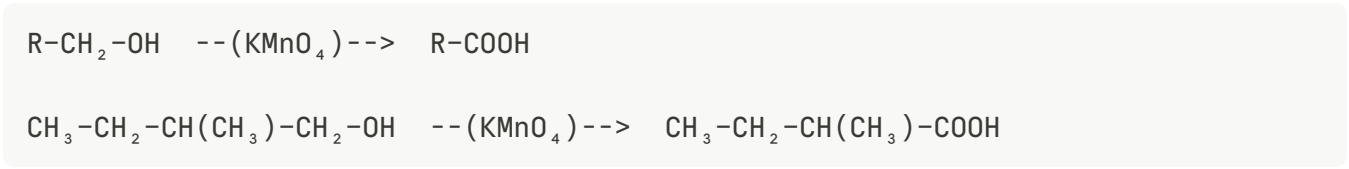


(b) Ring substitution in aromatic acids:

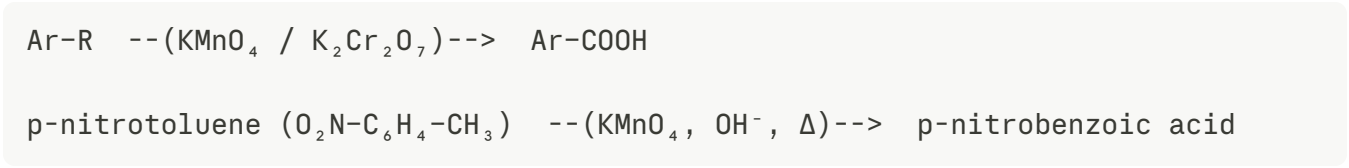


Preparation of carboxylic acids

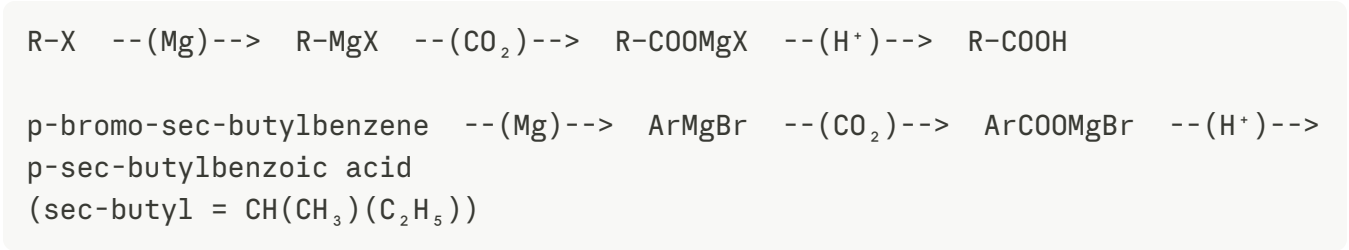
1. Oxidation of 1° alcohols



2. Oxidation of alkylbenzenes



3. Carbonation of Grignard reagents



4. Hydrolysis of nitriles

```
R–C≡N  +  H2O   --(acid or base)-->   R–COOH

benzyl chloride  --(NaCN)-->  C6H5–CH2–CN  --(75% H2SO4, reflux)-->  C6H5–
CH2–COOH
                              phenylacetonitrile
phenylacetic acid

n-butyl chloride  --(NaCN)-->  n-C4H9CN  --(aq. alc. NaOH, reflux)-->  n-
C4H9COO- + NH3  --(H+)-->  n-C4H9COOH + NH4+
                               n-valeronitrile (pentanenitrile)
valeric acid
```

5. Malonic ester synthesis

```
R–CH(COOC2H5)2   --(H2O, OH-, Δ)-->   R–CH(COO-)2   --(H+)-->   R–CH(COOH)2
--(Δ, 140 °C)-->   R–CH2–COOH  +  CO2
monoalkylmalonic ester
monosubstituted acetic acid
```

6. Special method for phenolic acids (Kolbe reaction)

```
sodium phenoxide (phenol salt)  +  CO2   --(125 °C, 4–7 atm)-->   sodium
salicylate (o-HO–C6H4–COONa)   --(H+)-->   salicylic acid (o-HO–C6H4–COOH)
```

9. Acid chlorides and acid anhydrides

Acid chlorides

Preparation: they are prepared from the corresponding acids by reaction with thionyl chloride, phosphorus trichloride or phosphorus pentachloride, as discussed in section 8.

Reactions

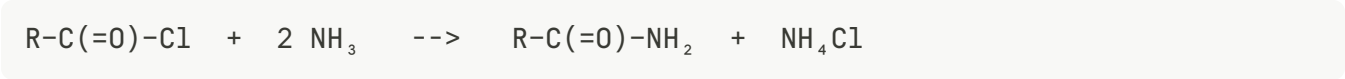
1. Conversion into acids and derivatives

```
R–C(=O)–Cl  +  H–Z   -->   R–C(=O)–Z  +  HCl          (written with "H2" in
the notes; read as H–Z)
```

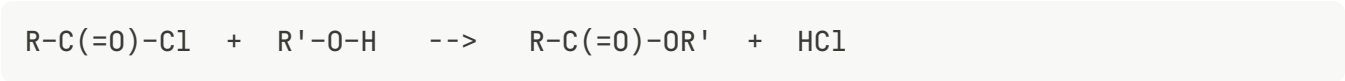
(a) Conversion into acids (hydrolysis):

```
R–C(=O)–Cl  +  H2O   -->   R–C(=O)–OH  +  HCl
```

(b) Conversion into amides (ammonolysis):



(c) Conversion into esters:

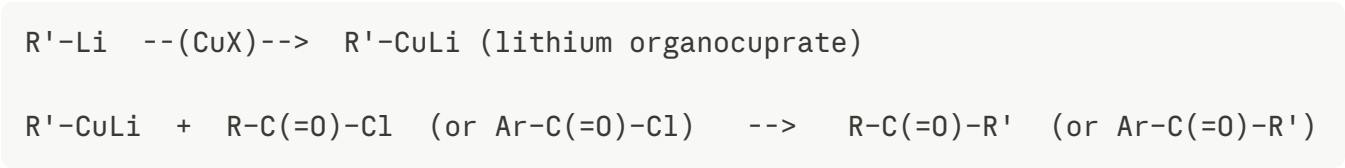


2. Formation of ketones

(a) Friedel–Crafts acylation:



(b) Reaction with organocopper compounds:



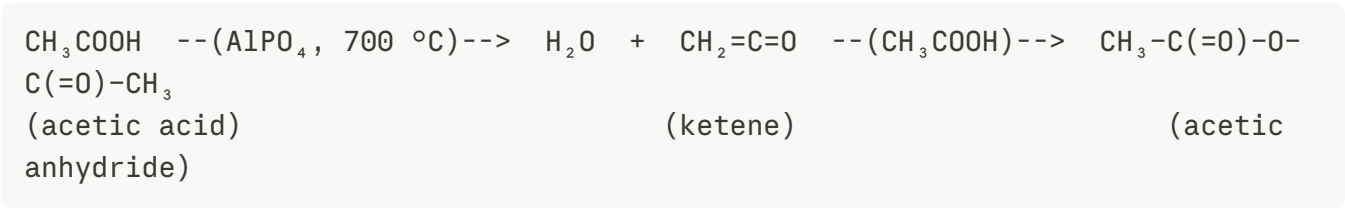
3. Formation of aldehydes by reduction



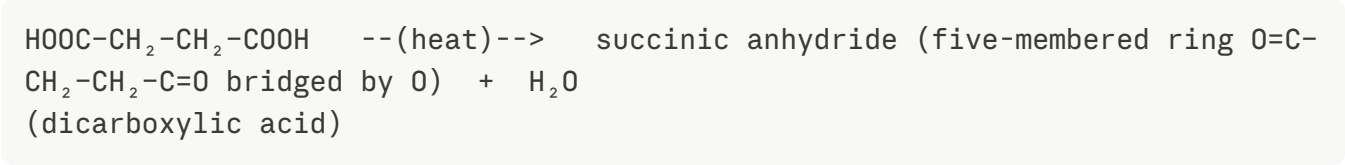
Acid anhydrides

Preparation

(i) From acetic acid via ketene:



(ii) From dicarboxylic acids:

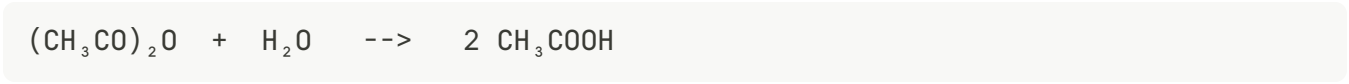


Reactions

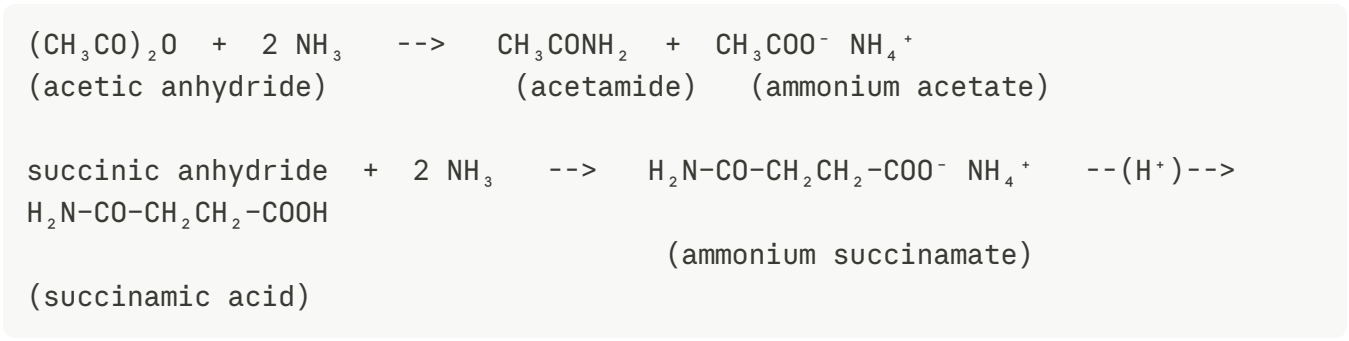
(i) Conversion into acids and acid derivatives



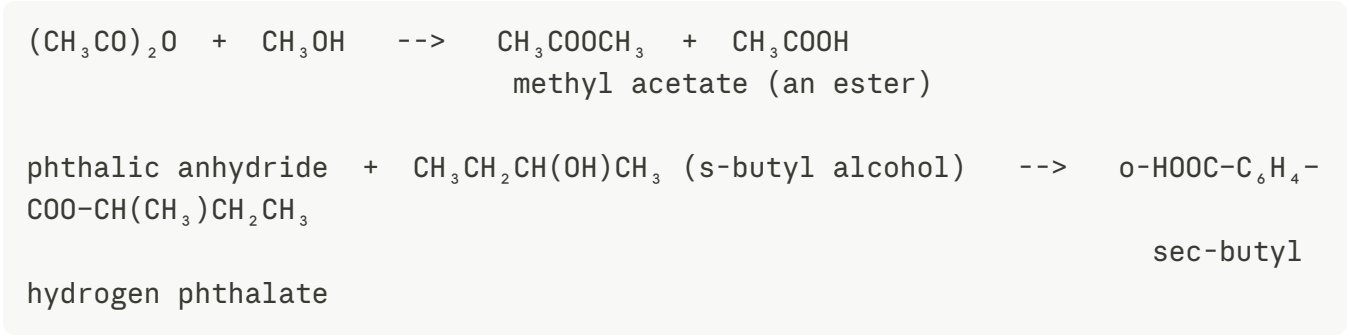
(a) Conversion into acids (hydrolysis):



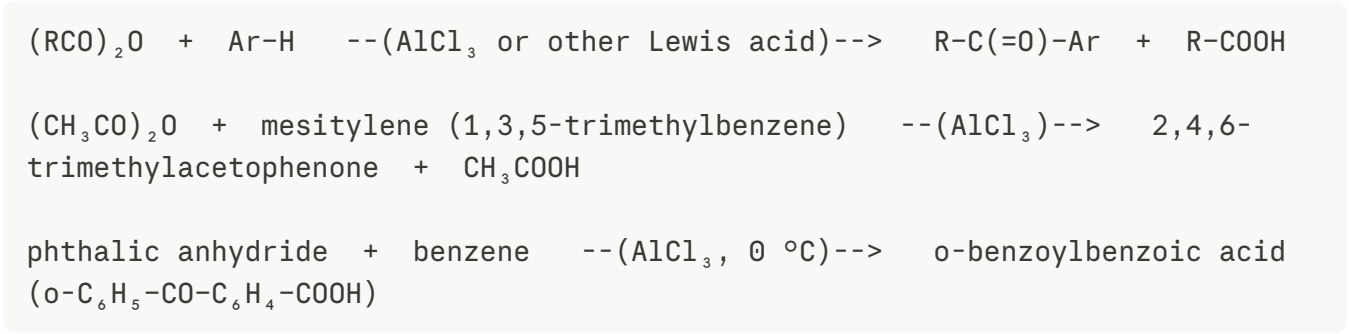
(b) Conversion into amides (ammonolysis):



(c) Conversion into esters (alcoholysis):



(ii) Formation of ketones (Friedel–Crafts acylation)



10. Amides and esters

Amides

Preparation

```
R–C(=O)–Cl  +  NH3   -->   R–C(=O)–NH2
```

Reactions

i) Hydrolysis

```
R–C(=O)–NH2  +  H2O   --(H+)-->   R–C(=O)–OH  +  NH4+
                      --(OH-)-->  R–COO-  +  NH3

benzamide (C6H5–CONH2)  +  H2SO4  +  H2O   -->   C6H5–COOH  +  NH4+ HSO4-
CH3CH2–CONH2  +  NaOH  +  H2O   -->   CH3CH2–COONa  +  NH3
```

ii) Conversion into imides

```
phthalic anhydride  --(2 NH3)-->  o-C6H4(CONH2)(COO- NH4+)  --(H+)-->  o-
C6H4(CONH2)(COOH)

phthalic anhydride  --(NH3, Δ)-------------------------------------->
phthalimide
o-C6H4(CONH2)(COO- NH4+)  --(Δ, 300 °C)-------------------------->
phthalimide
o-C6H4(CONH2)(COOH)  --(Δ)---------------------------------------->
phthalimide

phthalimide = benzene ring fused to a five-membered ring: C(=O)–NH–C(=O)
```

iii) Hofmann degradation

```
R–CONH2  (or Ar–CONH2)   --(OBr-)-->   R–NH2  (or Ar–NH2)  +  CO3 2-
```

Esters

Preparation

- **From acids** (reactivity of R′OH: 1° > 2° > 3°):

```
R–C(=O)–OH  +  R'O–H   ⇌ (H+)   R–C(=O)–OR'  +  H2O
```

- **From acid chlorides or anhydrides:**

```
R–C(=O)–Cl  +  R'–OH   -->   R–C(=O)–OR'  +  HCl
(RCO)2O  +  R'–OH   -->   R–COOR'  +  R–COOH
```

- **From esters (transesterification):**

```
R–C(=O)–OR'  +  R''O–H   ⇌ (H⁺ or ⁻OR'')   R–C(=O)–OR''  +  R'–OH
```

Mechanism of acid-catalysed transesterification (the notes title this "acid-catalyzed esterification")

1. Protonation of the carbonyl oxygen of Ester A, R–C(=O)–OR′, gives R–C(=O⁺H)–OR′ ↔ R–C⁺(OH)–OR′.
2. Alcohol B (R′′–OH) attacks the carbonyl carbon: R–C(OH)(OR′)–O⁺(H)R′′.
3. Proton transfer gives R–C(OH)(O⁺HR′)–OR′′.
4. Loss of Alcohol A (R′–OH) gives R–C(=O⁺H)–OR′′.
5. Loss of H⁺ gives **Ester B**, R–C(=O)–OR′′. Each step is reversible.

Mechanism of base-catalysed transesterification

```
R–C(=O)–OR'  +  ⁻OR''   ⇌   R–C(O⁻)(OR')(OR'')   ⇌   R–C(=O)–OR''  +  ⁻OR'
Ester A         alkoxide B     tetrahedral intermediate    Ester B
alkoxide A
```

Reactions of esters

Conversion into acids and derivatives

(a) Conversion into acids (hydrolysis):

```
R–C(=O)–OR'  +  H2O   --(H⁺)-->   R–C(=O)–OH  +  R'–OH
                      --(OH⁻)-->  R–COO⁻  +  R'–OH
```

(b) Conversion into amides:

```
R–C(=O)–OR'  +  NH3   -->   R–C(=O)–NH2  +  R'–OH
```

(c) Conversion into esters (transesterification):

```
R–C(=O)–OR'  +  R''–OH   ⇌ (acid or base)   R–C(=O)–OR''  +  R'–OH

glyceride (a fat):  CH2–O–C(=O)–R / CH–O–C(=O)–R' / CH2–O–C(=O)–R''   +
CH3OH   --(acid or base)-->
    R–COOCH3 + R'–COOCH3 + R''–COOCH3  (mixture of methyl esters)   +
HOCH2–CH(OH)–CH2OH  (glycerol)
```

Reaction with Grignard reagents

```
R–C(=O)–OR'  +  2 R''–MgX   -->   R–C(OH)(R'')₂        (3° alcohol)
```

Reduction to alcohols

- Catalytic hydrogenation (hydrogenolysis):

```
R–COOR'  +  2 H₂   --(CuO·CuCr₂O₄, 250 °C, 3000–6000 lb/in²)-->   R–CH₂OH  +
R'–OH
```

- Chemical reduction:

```
4 R–COOR'  +  2 LiAlH₄   --(anhydrous ether)-->   { LiAl(OCH₂R)₄  +
LiAl(OR')₄ }   --(H⁺)-->   { R–CH₂OH,  R'OH }
```

Reaction with carbanions: Claisen condensation

```
CH₃–C(=O)–OR'  +  (CH₃)₂CH–C(=O)–OR''   --(⁻OC₂H₅)-->   CH₃–C(=O)–C(CH₃)₂–
C(=O)–OR''
                                                          (a β-keto ester)
```